2.3 Homework: Constant Acceleration Solutions

1. Vernier cart. Knowns: $v_0 = 0$, $a = 0.500$ m/s^2, $x_0 = 0$.

   (a)

t (s)	0	1.00	2.00	3.00	4.00
v (m/s)	**0**	**0.500**	**1.00**	**1.50**	**2.00**

   Each second adds 0.500 m/s; this is $v = v_0 + at$ one row at a time. The 0.500 m/s at $t = 1.00$ s is the instantaneous velocity found in 2.2, question 4(b).

   (b) Want $x - x_0$ at $t = 2.00$ s. Use $x = x_0 + v_0 t + \frac{1}{2}at^2$:

   $x - x_0 = 0 + \frac{1}{2}(0.500 \text{ m/s}^2)(2.00 \text{ s})^2 = \mathbf{1.00\ m}$. **Yes, it agrees** with the measured 1.000 m.

   Also accept the average velocity route: $\bar{v} = \frac{1.00+0}{2} = 0.500$ m/s, and $0.500 \text{ m/s} \times 2.00 \text{ s} = 1.00$ m. The 2.2 data were made from $x = 0.25\,t^2$, which is exactly $\frac{1}{2}at^2$ with $a = 0.500$ m/s^2.

2. Tesla Model S Plaid. Knowns: $v_0 = 0$, $v = 26.8$ m/s, $t = 1.99$ s.

   (a) Want a. Use $v = v_0 + at$, so $a = \dfrac{v - v_0}{t}$:

   $a = \dfrac{26.8 \text{ m/s} - 0}{1.99 \text{ s}} = 13.467\ldots \approx \mathbf{13.5\ m/s^2}$

   The car gains 13.5 m/s of speed every second, more than a falling object's 9.80 m/s^2. Tesla's figure leaves out the first foot of travel, as drag-strip timing does, so the true average acceleration from a standstill is a little lower.

   (b) Want $x - x_0$. Use $\bar{v} = \dfrac{v + v_0}{2}$: $\bar{v} = \dfrac{26.8 + 0}{2} = 13.4$ m/s.

   $x - x_0 = \bar{v}\,t = 13.4 \text{ m/s} \times 1.99 \text{ s} = 26.666 \approx \mathbf{26.7\ m}$

   Also accept $x - x_0 = \frac{1}{2}at^2 = \frac{1}{2}(13.467\ldots)(1.99)^2 \approx 26.7$ m. Using the rounded 13.5 m/s^2 gives 26.73 m, which also rounds to 26.7 m.

3. Braking. Knowns: $v_0 = 13.4$ m/s, $v = 0$, $x - x_0 = 14$ m. Take the direction of travel as positive.

   (a) Want a; no time is given. Use $v^2 = v_0^2 + 2a(x - x_0)$:

   $0 = (13.4 \text{ m/s})^2 + 2a(14 \text{ m})$, so

   $a = -\dfrac{(13.4 \text{ m/s})^2}{2(14 \text{ m})} = -\dfrac{179.56}{28} = -6.4128\ldots \approx \mathbf{-6.4 \text{ m/s}^2}$

   It is negative because the car is slowing down: the acceleration points backward, opposite to the forward velocity.

   Two significant figures, because 14 m has two. Accept -6.41 m/s^2. A positive 6.4 m/s^2 described as "deceleration" earns the number but not the sign question.

   (b) Knowns: $v_0 = 26.8$ m/s, $v = 0$, $a = -6.4128\ldots$ m/s^2. Want $x - x_0$. Use $v^2 = v_0^2 + 2a(x - x_0)$ again:

   $x - x_0 = \dfrac{v^2 - v_0^2}{2a} = \dfrac{0 - (26.8 \text{ m/s})^2}{2(-6.4128\ldots \text{ m/s}^2)} = \dfrac{718.24}{12.8257\ldots} = \mathbf{56 \text{ m}}$

   Twice the speed gives four times the braking distance: 4×14 m $= 56$ m.

   The distance grows with v_0^2, and $26.8 = 2 \times 13.4$, so v_0^2 is four times as large. The Highway Code prints 55 m; its table is rounded, and with two significant figures 55 m and 56 m agree. Accept 56.1 m from the rounded -6.4 m/s^2.

4. Shanghai Tower elevator. Knowns: $v_0 = 0$, $v = 20.5$ m/s, $a = 1.0$ m/s^2.

   (a) Want t. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:

   $t = \dfrac{20.5 \text{ m/s} - 0}{1.0 \text{ m/s}^2} = 20.5 \approx \mathbf{21 \text{ s}}$

   Accept 20.5 s. The acceleration has two significant figures, so 21 s is the strict answer; either earns the mark.

   (b) Want $x - x_0$. Use $\bar{v} = \dfrac{v + v_0}{2} = \dfrac{20.5 + 0}{2} = 10.25$ m/s:

   $x - x_0 = \bar{v}\, t = 10.25 \text{ m/s} \times 20.5 \text{ s} = 210.125 \approx \mathbf{210 \text{ m}}$

   Also accept $v^2 = v_0^2 + 2a(x - x_0)$: $x - x_0 = \frac{(20.5)^2}{2(1.0)} = 210.125$ m, or $\frac{1}{2}at^2 = \frac{1}{2}(1.0)(20.5)^2$. Carry 20.5 s, not the rounded 21 s, into this step.

   (c) Speeding up takes 210 m and slowing down takes another 210 m: 2×210.125 m $= 420.25 \approx \mathbf{420 \text{ m}}$.

   On any ride shorter than about 420 m the elevator must start slowing down before it reaches 20.5 m/s. The tower is 632 m tall, so a long ride such as the express from the lobby to the observation deck gets there; short hops between floors never do.